# A3-PS016-TC012

## Simulation-Based Handover Decision Evaluation for High-Mobility Users

AIORI Hackathon 3 | 6G & Future Networks


MENTORED BY

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# Problem Description

## Research problem

High-speed user mobility makes reliable cellular handover difficult. A late decision can interrupt connectivity; an unnecessary reversal can cause ping-pong switching. The problem is to determine whether a supervised, mobility-aware handover policy performs differently from a conventional signal-based policy under controlled LTE simulation conditions.

## Challenge and mandatory evaluation scope

- Benchmark A3-RSRP against one supervised Random Forest decision model in ns-3 LTE.
- Configure three cells, two mobility speeds, and multiple random seeds.
- Measure handover failures, ping-pong events, interruption duration, and throughput.
- Produce repeatable experiments, comparisons, visualizations, and a final technical report.

# Background

## Mobility management in cellular networks

LTE and 5G cellular systems maintain service as users move between coverage areas. When a mobile user equipment (UE) approaches a new cell, the network must decide whether and when to transfer the active connection. Poor timing can cause service interruption, unsuccessful transfers, or rapid back-and-forth handovers.

## Existing approach and motivation

Conventional LTE handover commonly relies on reference signal received power (RSRP), hysteresis and time-to-trigger (TTT). These mechanisms are standardized and effective, but a fixed configuration can have different trade-offs across mobility speeds and changing radio conditions. Supervised learning may use additional information, such as user speed and recent signal trends, to support context-aware decisions.

## Project direction

This work proposes a reproducible ns-3 LTE comparison of the conventional A3-RSRP policy with one Random Forest policy, evaluated using three cells, two mobility speeds, and multiple seeds. The study will test performance differences; it will not assume ML superiority or claim 6G conformance.

# Literature & RFC Review

## Selected standards, tools and research

| | |
|---|---|
| **3GPP TS 36.331 [1]** | Defines LTE RRC Event A3 measurement reporting and associated offsets, hysteresis, and time-to-trigger. Grounds the conventional policy. |
| **ns-3 LTE model [2], [3]** | Provides the A3-RSRP handover algorithm and simulation tracing interfaces. Grounds practical baseline implementation. |
| **Mollel et al. [4]** | Surveys ML techniques for cellular handover management and discusses mobility-performance challenges. |
| **Wang et al. [5]** | Studies supervised mobility prediction, including Random Forest, in reducing unnecessary handovers in its own simulated scenario. |
| **IETF RFC 2330 [6]** | Framework for IP performance metrics: precise definitions, repeatable methods, measurement uncertainty and statistical interpretation. |

Research direction. Evaluate whether a mobility-aware Random Forest decision policy changes reliability, interruption, ping-pong behavior and throughput relative to A3-RSRP under matched ns-3 conditions. RFC 2330 guides measurement discipline, not LTE handover signaling.

# Proposed Solution

## Mobility-Aware Supervised Handover Benchmarking Framework

The proposal uses ns-3 LTE with three eNodeBs and a mobile UE. A conventional A3-RSRP handover algorithm establishes the reference policy, while a supervised Random Forest classifier predicts a stay / handover decision from measurements available at decision time.

| | | |
|---|---|---|
| 01 | **Scenario design** | Three cells; two speeds; traffic loads and deterministic seeds. |
| 02 | **Data and labels** | Collect serving/neighbor RSRP, speed and signal trend; derive training labels from a documented offline simulation-outcome criterion. |
| 03 | **Model training** | Train and validate one Random Forest on training scenarios; reserve independent evaluation scenarios and seeds. |
| 04 | **Policy execution** | Evaluate A3-RSRP and trained ML policy in separate runs. ML handover commands use a custom ns-3 integration layer. |
| 05 | **Benchmark and report** | Compute four required metrics, aggregate results by speed and seed, then visualize uncertainties and trade-offs. |

Scope: This is a proposed integration plan. No simulation performance improvement is claimed prior to experimental validation.

# Optimization Proposed

The proposed enhancement tests context-aware handover decisions while preserving a fair, signal-based A3-RSRP baseline. Benefits listed below are optimization targets, not measured gains.

| Dimension | Before: conventional A3-RSRP | After: proposed Random Forest |
|---|---|---|
| Decision logic | RSRP-based Event A3; fixed hysteresis and TTT | Supervised action prediction from multiple inputs |
| Mobility awareness | Speed is not a direct decision feature | Includes current UE speed |
| Signal processing | Current/filtered RSRP measurements | RSRP difference plus recent signal trends |
| Switching stability | Hysteresis and time-to-trigger | Confidence gate / cooldown investigated as safeguards |
| Failure / ping-pong | Measured baseline outcomes | Seek reductions; verify on held-out runs |
| Interruption / throughput | Measured baseline service outcomes | Seek lower interruption without throughput loss |
| Assessment | Reference policy results | Same topology, seeds and traffic; statistical comparison |

Optimization objective: determine the conditions in which a learned policy improves handover reliability and service continuity, and where the added safeguards or prediction errors create trade-offs.

# Solution Architecture & Design

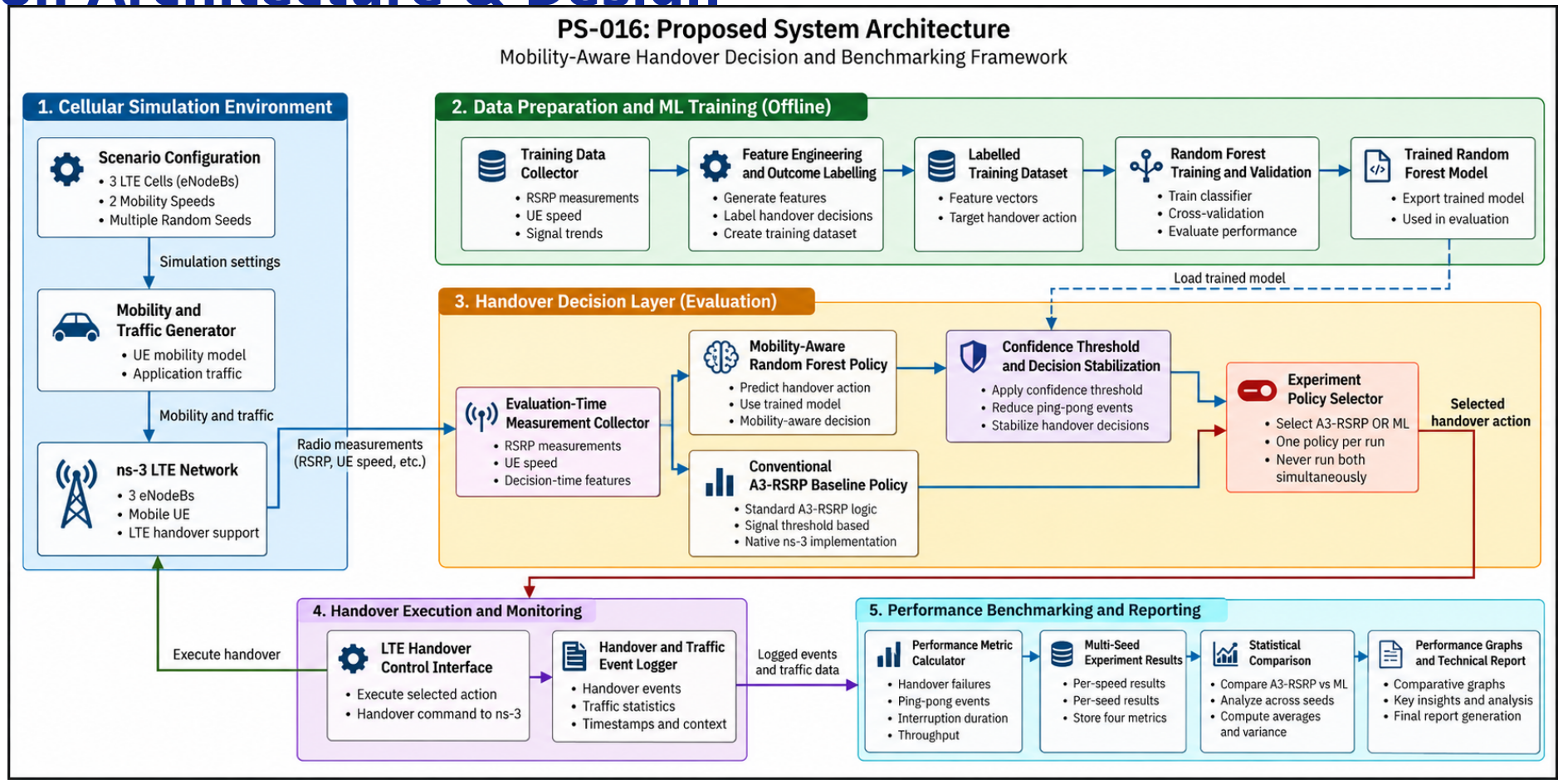



Figure 1. Proposed five-layer architecture: ns-3 simulation, offline data preparation and training, policy evaluation, handover execution, and multi-seed performance reporting. A3-RSRP and the learned policy control separate experiments.

# Data Flow Diagram

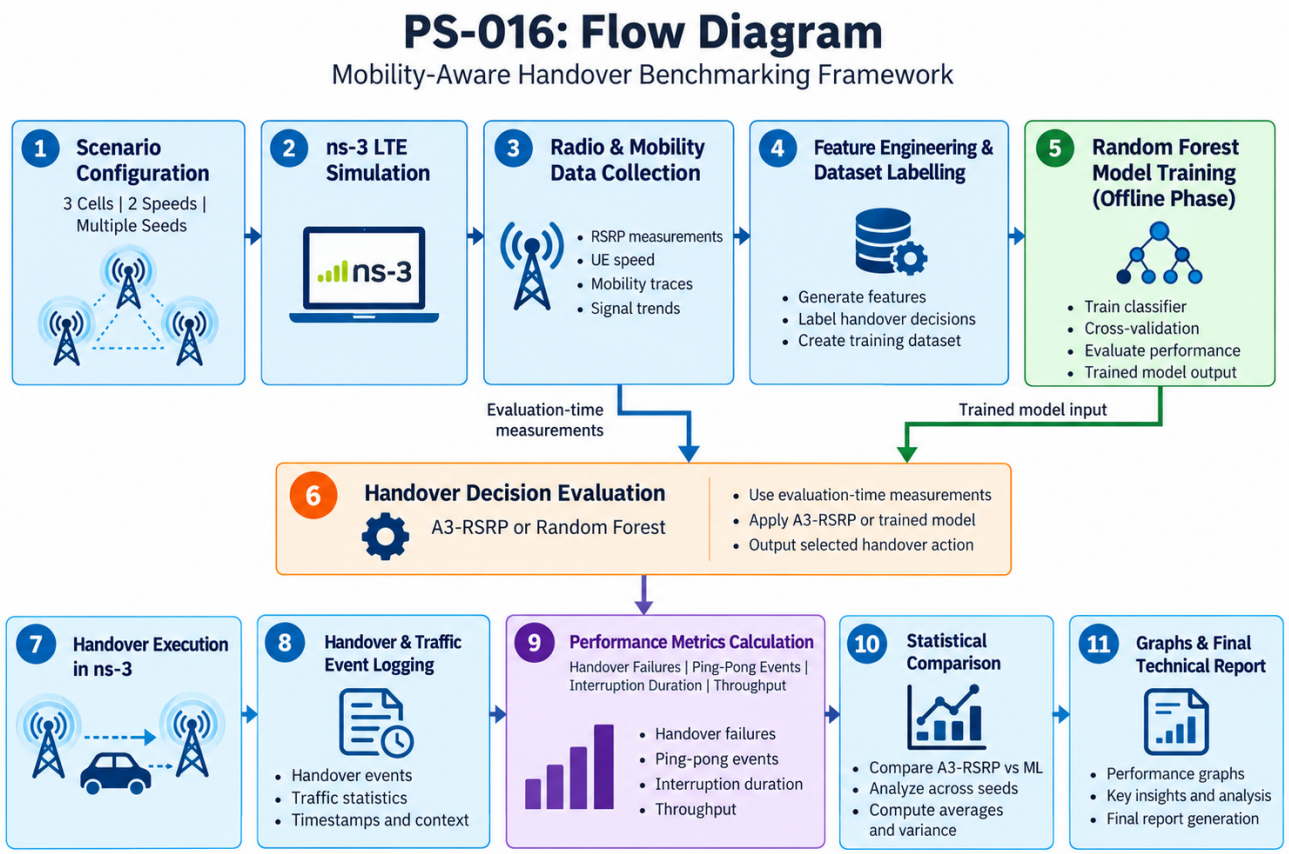



**Figure 2:** Flow Diagram of the Proposed Mobility-Aware Handover Benchmarking Framework

Figure 2. Proposed flow from configured ns-3 scenarios through offline training and independent handover-policy evaluation to event logging, metric calculation, statistical comparison and the technical report.

# Deployment Timeline

Planned development schedule: 10-30 October 2026. Activities describe the intended post-proposal implementation; the Level 1 PDF/video are separate submission tasks.

| Dates | Milestone | Method / completion checkpoint |
|---|---|---|
| 10-12 Oct | Environment | Install ns-3 LTE; validate an example and handover tracing. |
| 13-15 Oct | Cellular scenario | Configure three cells, two UE speeds, traffic and seed control. |
| 16-18 Oct | A3-RSRP baseline | Configure hysteresis / TTT; log handover events and traffic. |
| 19-21 Oct | Offline supervised ML | Label controlled scenarios; train and validate Random Forest. |
| 22-24 Oct | ML integration | Connect predictions to handover requests; verify one policy per run. |
| 25-27 Oct | Experiments | Run paired multi-seed evaluations; compute four mandatory metrics. |
| 28-30 Oct | Analysis / report | Review variability, create plots, verify reproducibility, demonstrate. |

Methodology: iterative validation, held-out evaluation scenarios, matched network conditions, documented seeds, and consistent metric definitions guided by RFC 2330. Major risks are training-label design and custom ML control integration.

# Bibliography


**[1]** 3GPP, TS 36.331: E-UTRA Radio Resource Control (RRC); Protocol Specification, Event A3, Sec. 5.5.4.4. https://www.3gpp.org/dynareport/36331.htm

**[2]** ns-3 Project, LTE Module - Model Design Documentation. https://www.nsnam.org/docs/models/html/lte-design.html

**[3]** ns-3 Project, LTE Module - User Documentation. https://www.nsnam.org/docs/models/html/lte-user.html

**[4]** M. S. Mollel et al., “A Survey of Machine Learning Applications to Handover Management in 5G and Beyond,” IEEE Access, vol. 9, 2021. DOI: 10.1109/ACCESS.2021.3067503.

**[5]** D. Wang et al., “Mitigating Unnecessary Handovers in Ultra-Dense Networks through Machine Learning-based Mobility Prediction,” arXiv:2302.11878, 2023. https://arxiv.org/abs/2302.11878

**[6]** V. Paxson, G. Almes, J. Mahdavi and M. Mathis, “Framework for IP Performance Metrics,” IETF RFC 2330, May 1998. https://www.rfc-editor.org/rfc/rfc2330.html

# Thank You for Your Attention!

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